

Anuk Centellas

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Computational linguist interested in applying ML and NLP to language-focused research and engineering problems.

EDUCATION

University of Washington

MS in Computational Linguistics

Sept 2025 – Present

4.0 GPA, Expected Graduation: December 2027

Western Washington University

BS in Mathematics/Computer Science, BA in Linguistics

Sept 2021 – June 2025

4.0 GPA, Honors Interdisciplinary Studies Minor, Outstanding Graduate in Linguistics

Coursework: Machine Learning, NLP, Statistics and Probability, Algorithms and Data Structures, Advanced Statistical Methods for NLP

EXPERIENCE

Language Revitalization Tooling Research

Feb 2026 – Present

Bridging Digital Tools for Linguistic Documentation and Revitalization

University of Washington

- Collaborated with student researchers to create langlit, a language revitalization app that addresses previous versatility gaps
- Performed user testing and documentation to improve organization, built-in instruction, and overall user-friendliness of langlit
- Created comprehensive tutorials for langlit users of different backgrounds, including documentary linguists, teachers, and learners
- Co-authored a paper titled "Bridging Digital Tools for Linguistic Documentation and Revitalization" published in **AmericasNLP**

Research for Intelligence & Security Internship

June 2026 – August 2026

LLM-Powered Content Collection Tool - National Foreign Language Center (NFLC)

University of Maryland

- Built an LLM pipeline from scratch to collect lesson content in 7 languages, reducing manual collection time from hours to minutes
- Refactored codebase for optimal modularity and organization, ensuring future work can be integrated smoothly
- Presented updates to other interns and leaders throughout the internship and addressed their questions and feedback
- Implemented third-party APIs for LLM calling, web searching, text scraping, and audio transcription
- Applied in-context learning and prompt engineering to boost LLM performance on classification and query generation
- Created a Gradio UI with instructions for entering desired content specifications to maximize user-friendliness

Language Revitalization Lab

Feb 2025 – June 2026

Documentation, Analysis, and Revitalization in Linguistics (DARLing) - Dr Maura O'Leary

Western Washington University

- Analyzed and structured under-documented linguistic data for integration into a morphological transducer system
- Improved code consistency and correctness by identifying and correcting errors affecting downstream translation accuracy
- Collaborated with the team to determine representations for ambiguous verb morphology that prioritize the speaker community

Machine Learning Research

April 2024 – June 2025

Hutch Research - Dr Brian Hutchinson

Western Washington University

- Collaborated with the Speech Pathology Department on an applied ML research project to transcribe audio of toddler speech
- Built an end-to-end data preprocessing pipeline in Python to transform noisy transcripts into structured, model-ready datasets
- Independently researched and tested existing forced alignment and diarization models to identify components for our system
- Communicated weekly with non-technical stakeholders to ensure our project was maximally useful for speech pathologists
- Coordinated task assignments, communication, and collaborative research sessions to ensure project milestones were met

PROJECTS

Data Formatting and Plotting App

Aug 2024 – June 2025

Language Learning Research Group (LLRG)

Python, Streamlit, pandas

- Built and deployed a Streamlit web application enabling users to upload raw Excel data and generate structured outputs
- Collaborated with a non-technical team and users to design the app, and independently learned Streamlit to implement it
- Designed and implemented a backend data normalization pipeline using pandas to standardize inconsistent formatting
- Generated exportable visualizations and reformatted data used by the LLRG to improve the Spanish program
- Presented the app and design process to a diverse, non-technical audience as my Honors Senior Capstone Project

Phoneme Classifier

Nov 2024

Deep/Machine Learning Project

Python, Scikit-learn, WandB, GitHub

- Trained and evaluated an ensemble model of twelve neural network classifiers on 1.7 million training instances
- Improved ensemble model performance by giving the top four performing NNs more votes
- Conducted extensive hyperparameter tuning and compared performance across model architectures to find the optimal model
- Achieved a validation accuracy of 0.456 and a test accuracy z-score of 1.340, the top performance for this task

Arbitrarily Deep Neural Network

Oct 2024

Deep/Machine Learning Project

Python, NumPy, argparse, GitHub

- Built a fully customizable neural network framework from scratch through pair programming
- Implemented support for both classification and regression tasks, minibatching, and an optional verbose mode
- Designed the neural network to accept data files and hyperparameters as command-line arguments for flexible experimentation
- Incorporated periodic evaluation on the development set with real-time performance reporting

TECHNICAL SKILLS

Python, Java, SQL; NumPy, pandas, Scikit-learn, Matplotlib, argparse, Weights & Biases; Git, GitHub, Linux/Unix, Streamlit, Gradio; data preprocessing, model evaluation, hyperparameter tuning, experiment tracking, LLM and NLP methods